Neural Codecs as Biosignal Tokenizers

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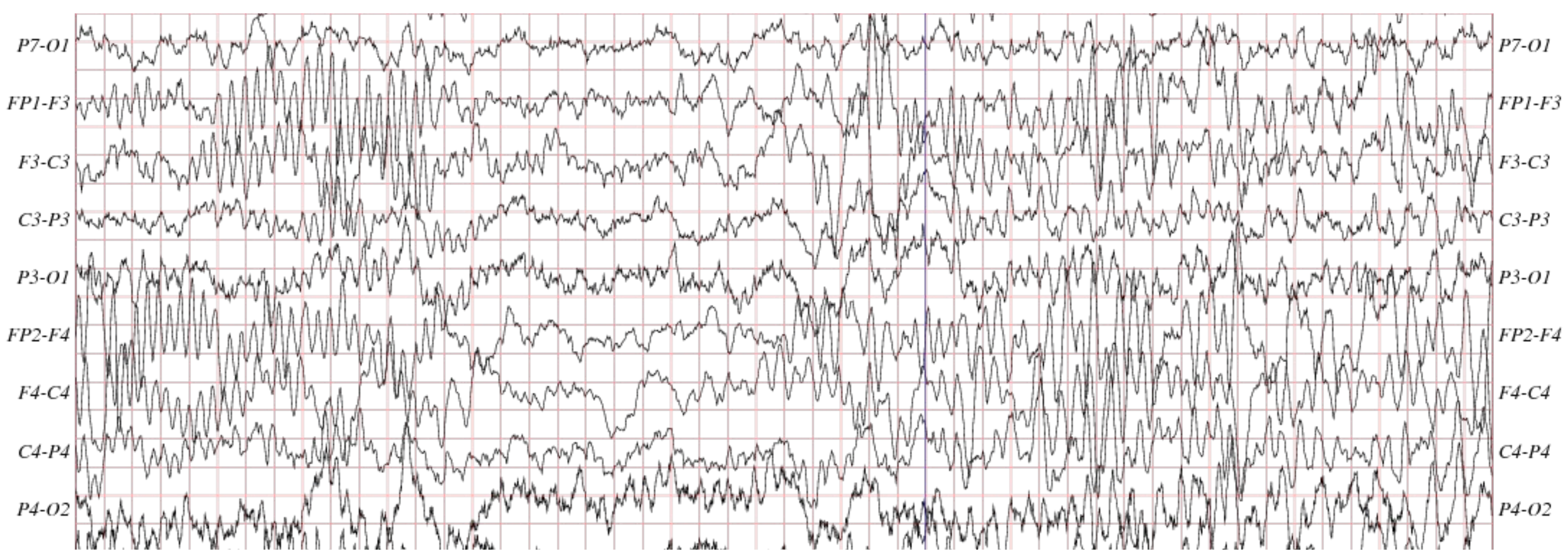
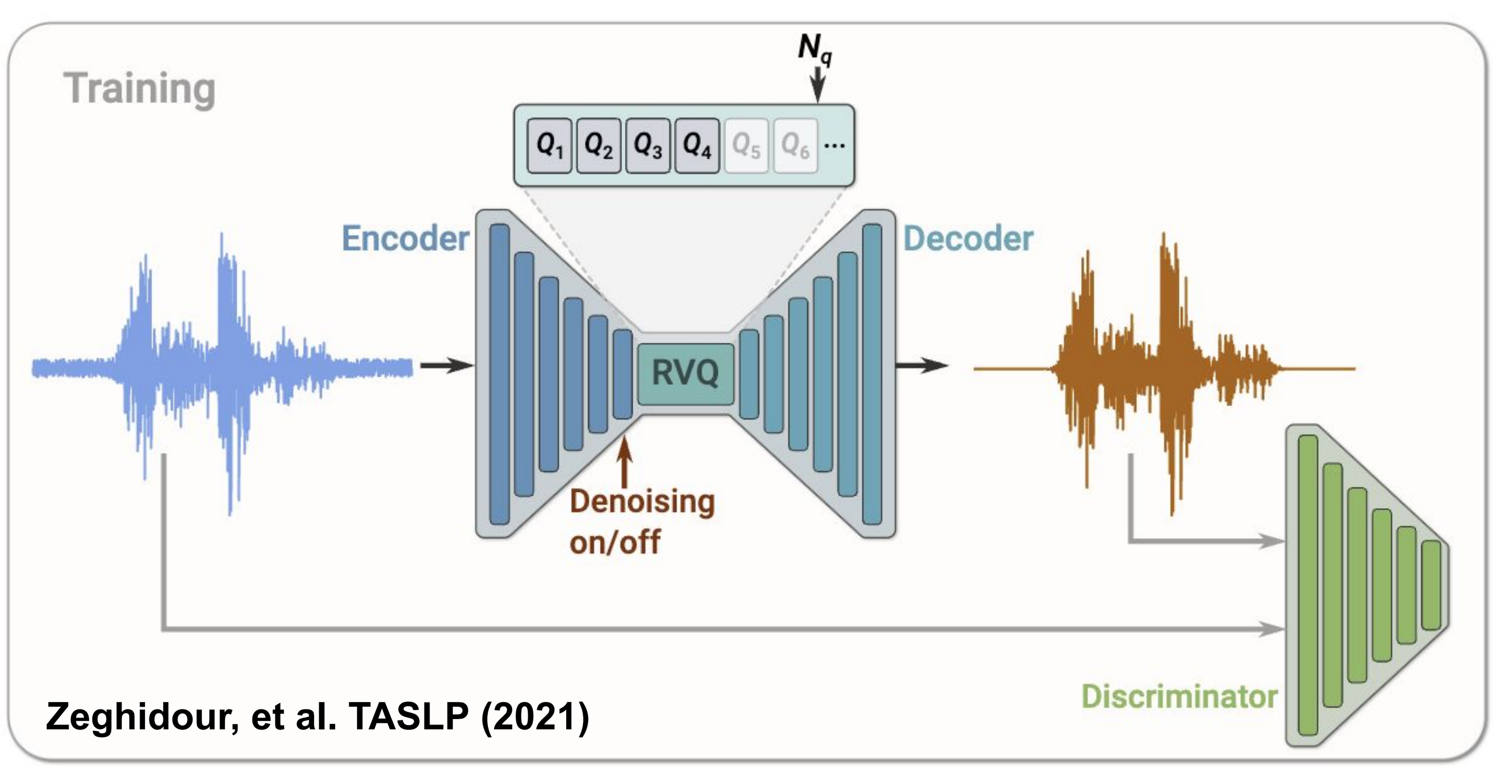
Biosignal Modalities: Noisy and heterogeneous representations



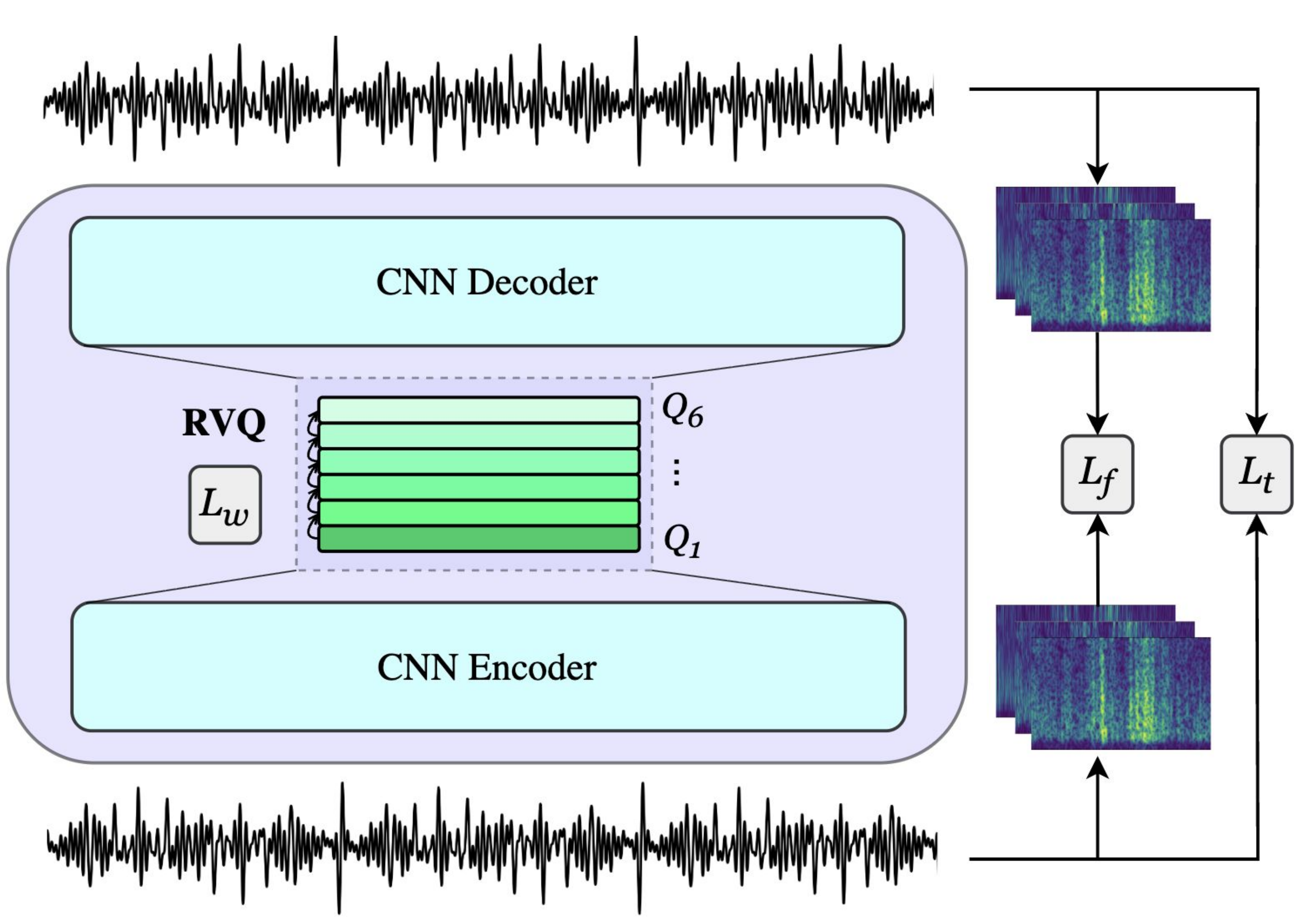
→ Sparse signal information should be **distilled** from the dense, multivariate input complexity
→ Can we impose an information bottleneck in a foundation model of biosignal modalities?

Proposal: Vector Quantization



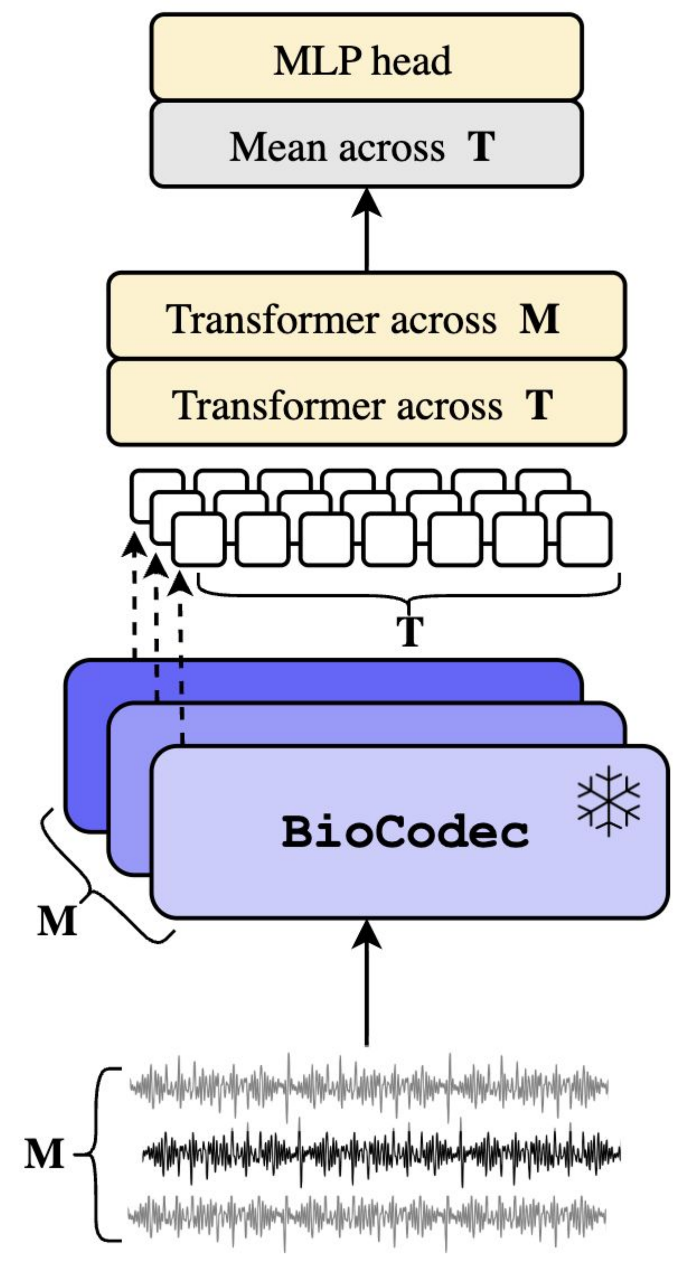
→ Discrete tokens model **structured dynamics** that are repeatable across the training set
→ Residual Vector Quantization (**RVQ**): Enables pattern hierarchies and multiscale modeling

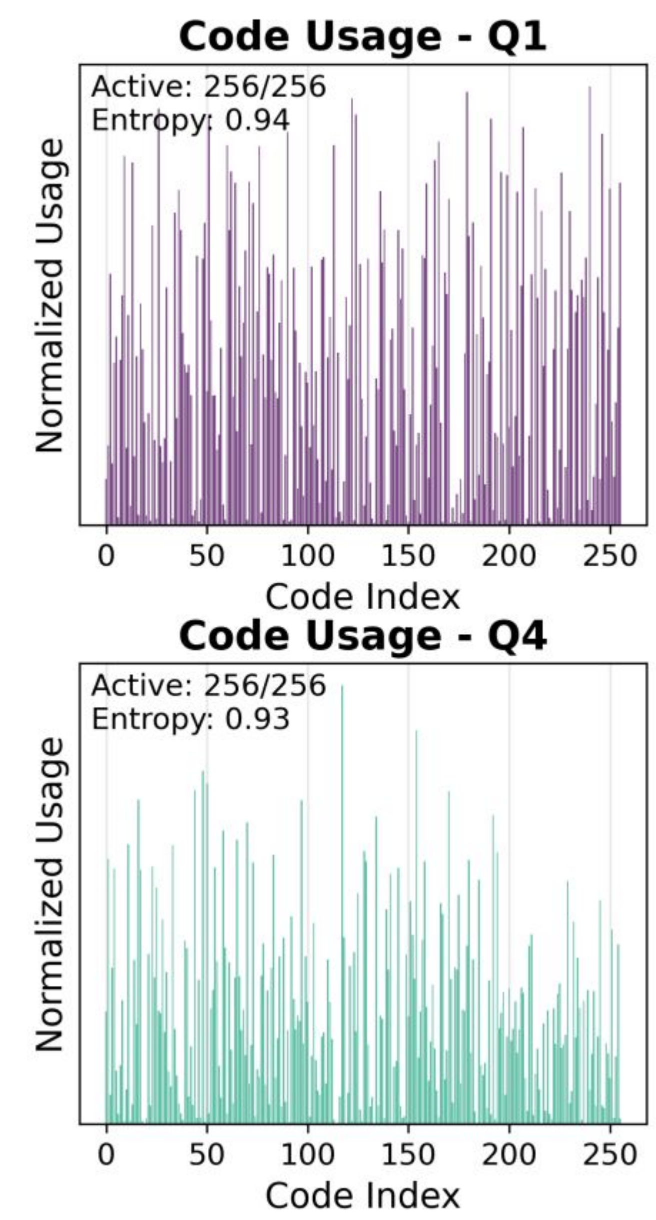
BioCodec: The architecture

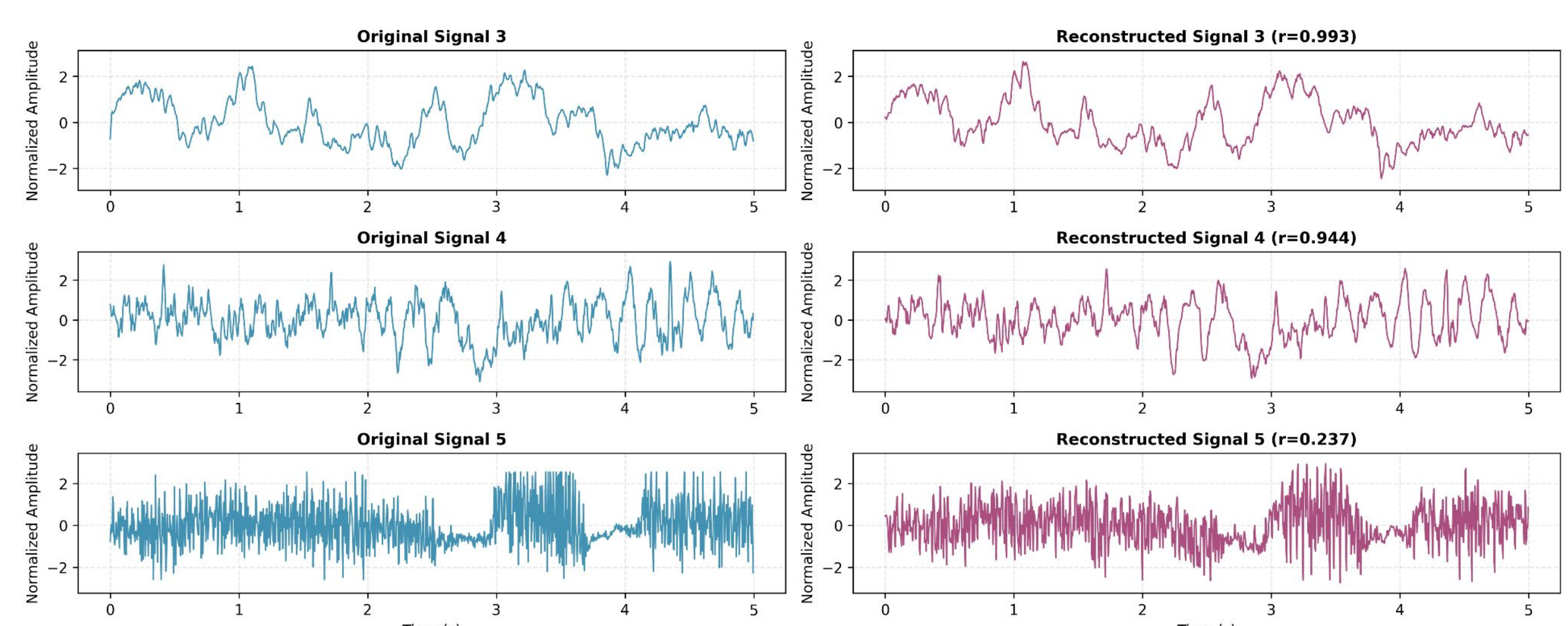


→ We pre-train a biosignal foundation model on reconstructing **single-channel** input recordings

BioCodec: Sanity Checks



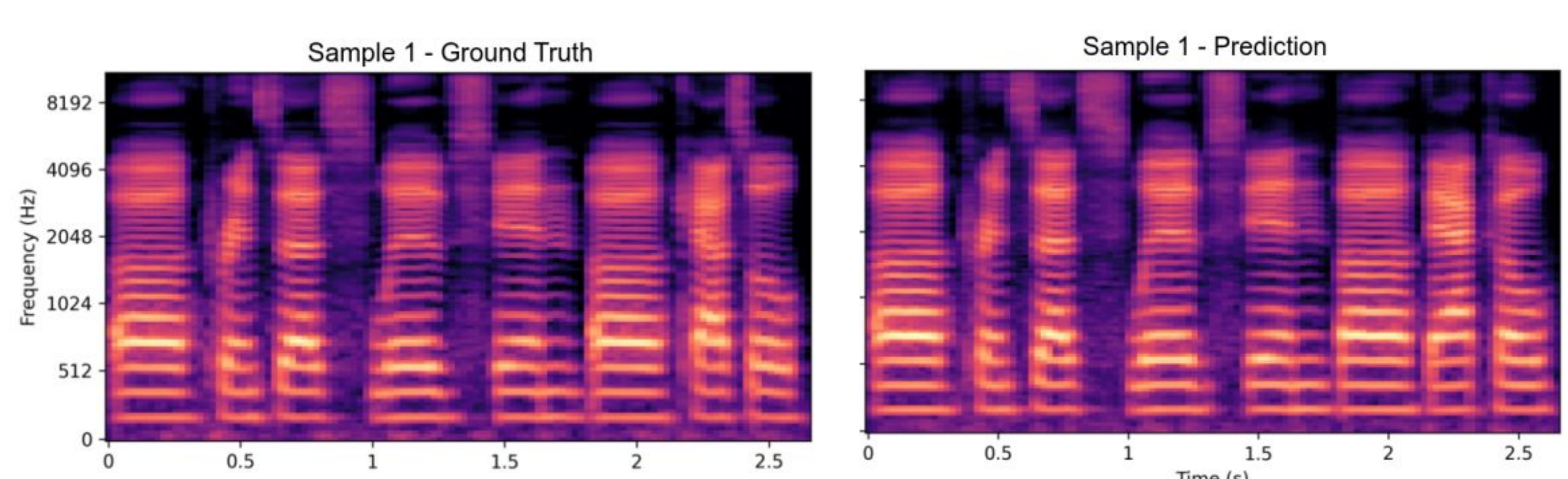




BioCodec: EEG Downstreams

Clinical

Methods	Model size	BAC	AUPRC	AUROC
SPaRCNet	0.8M	0.790 ± 0.002	0.841 ± 0.002	0.868 ± 0.001
ContraWR	1.6M	0.775 ± 0.004	0.842 ± 0.010	0.846 ± 0.007
BIOT	3.2M	0.796 ± 0.006	0.879 ± 0.002	0.882 ± 0.004
EEGPT	4.7M	0.796 ± 0.002	0.897 ± 0.002	0.872 ± 0.001
LaBraM	5.8M	0.814 ± 0.002	0.897 ± 0.002	<u>0.902</u> ± 0.001
CBraMod	4.0M	**0.825** ± 0.001	**0.922** ± 0.001	**0.916** ± 0.002
BioCodec	1.0M	<u>0.816</u> ± 0.003	<u>0.907</u> ± 0.003	0.883 ± 0.002



It works for **EMG** too!

Dataset	Method	Model Size	BAC	Weighted F1
Ninapro DB2	TimesNet	2.3M	0.432 ± 0.001	0.835 ± 0.001
	BioCodec	0.7M	**0.542** ± 0.016	**0.849** ± 0.016
Ninapro DB3	TimesNet	2.3M	0.157 ± 0.034	**0.684** ± 0.010
	BioCodec	0.7M	**0.175** ± 0.062	0.595 ± 0.113
Ninapro DB5	TimesNet	2.4M	0.376 ± 0.027	0.813 ± 0.003
	BioCodec	0.8M	**0.453** ± 0.028	**0.817** ± 0.014
EMG MCS	TimesNet	2.3M	0.390 ± 0.032	0.365 ± 0.039
	BioCodec	0.4M	**0.606**± 0.028	**0.600** ± 0.032